Module 1: Introduction to Geometric Learning

Section 1.4 — Gauge Theories and Local Geometries

Data Science Master's Programme

Contents

Motivation

In Sections 1.1–1.2 we discussed *global* symmetries: a single group acts uniformly on all of $\mathbb{R}^D$. But what if the data lives on a *curved* surface? On a sphere, for instance, "North" means something different at different points. There is no globally consistent choice of orientation.

This is the starting point of *gauge theory*: the symmetry group acts *locally*, independently at each point of a manifold, and the key challenge is to define consistently how features at different points relate to each other.

Although the full mathematical theory involves differential geometry, we will develop the ideas at an accessible level, focusing on intuition and the connection to machine learning.

1 The Limitation of Global Symmetry

Recall from Section 1.2 that SO(2)-equivariant maps on $\mathbb{R}^2$ transform all feature vectors by the *same* rotation R_θ everywhere. This is appropriate when the domain is flat (Euclidean).

Now suppose the domain is a sphere S^2 (think of climate data defined at every point on Earth). At the North Pole, "East" points in a certain direction; at the South Pole, the same label "East" points in the opposite direction. There is no single global rotation that captures both simultaneously.

Remark 1.1: The need for local symmetry

On a curved manifold, we cannot define a *global* coordinate system that is consistent everywhere. Instead, we can only define *local* coordinate systems (called *gauges*) in small neighbourhoods, with the freedom to choose them independently at each point. The physics-inspired study of how these local choices relate to each other is called *gauge theory*.

Remark 1.2: Historical perspective

Gauge theory was developed by physicists (Yang, Mills, Weyl, and others) to formalise the freedom to choose local reference frames in field theory. Maxwell's equations, the Standard Model of particle physics, and general relativity are all gauge theories. The key insight — that physically meaningful quantities must be independent of the (arbitrary) choice of local frame — was imported into machine learning in the late 2010s as researchers extended convolutional networks beyond flat Euclidean domains.

2 Tangent Spaces and Local Frames

Definition 2.1: Tangent Space

Let $\mathcal{M}$ be a smooth d-dimensional manifold and $p \in \mathcal{M}$. The **tangent space** $T_p\mathcal{M}$ is the d-dimensional vector space of all velocity vectors of smooth curves passing through p. Intuitively, $T_p\mathcal{M}$ is the best linear (flat) approximation to $\mathcal{M}$ at p.

Example 2.1: Tangent space to the sphere

Let $\mathcal{M} = S^2$ and p be the North Pole. The tangent space T_pS^2 is the horizontal plane touching the sphere at the top — a copy of $\mathbb{R}^2$. At any other point $q \in S^2$, the tangent space T_qS^2 is another copy of $\mathbb{R}^2$, but oriented differently in ambient $\mathbb{R}^3$.

A **local frame** (or *gauge*) at p is a choice of basis $\{e_1(p), \ldots, e_d(p)\}$ for $T_p\mathcal{M}$. This is like choosing local coordinate axes. The freedom to choose a different frame at each point is the *gauge freedom*.

Example 2.2: Longitude/latitude as a gauge

On the Earth's surface, longitude and latitude define a local frame at each point (East and North directions). This frame breaks down at the poles (a well-known issue with Mercator projections) — a reminder that no single gauge is globally valid on S^2.

Remark 2.1: The hairy ball theorem

The impossibility of a globally consistent frame on S^2 is not an artefact of poor choices — it is a topological obstruction. The **hairy ball theorem** states that any continuous tangent vector field on S^2 must vanish at some point. In other words, you cannot "comb the hair on a coconut" without creating at least one cowlick. Equivalently: no smooth global frame exists on S^2. This forces gauge-theoretic methods on the sphere; no amount of cleverness in coordinate choice can sidestep the topology.

3 Fiber Bundles

To handle features defined at every point of a manifold, we need the concept of a *fiber bundle*.

Definition 3.1: Fiber Bundle (intuitive)

A **fiber bundle** E over a base manifold $\mathcal{M}$ with fiber F is a space that locally looks like $\mathcal{M} \times F$, but may be globally "twisted." There is a projection map $\pi : E \to \mathcal{M}$ such that for each $p \in \mathcal{M}$, the *fiber over* p is $\pi^{-1}(p) \cong F$.

Key examples:

- **Tangent bundle** $T\mathcal{M}$: the fiber over p is the tangent space $T_p\mathcal{M} \cong \mathbb{R}^d$. A *vector field* on $\mathcal{M}$ is a section of $T\mathcal{M}$: it assigns a vector in $T_p\mathcal{M}$ to each p.
- **Feature bundle**: in a geometric neural network, the fiber over each point p is the *feature vector space* $\mathbb{R}^c$ for that point. The *structure group* G acts on each fiber, encoding the local symmetry.
- **Möbius strip**: a non-trivial fiber bundle over S^1 with fiber $\mathbb{R}$ — the twist makes it globally non-orientable, illustrating how bundles can differ from simple products.

Definition 3.2: Section of a bundle

A **section** of a fiber bundle $\pi : E \to \mathcal{M}$ is a smooth map $s : \mathcal{M} \to E$ such that $\pi \circ s$ is the identity on $\mathcal{M}$. Equivalently: at every point p, the section picks out an element $s(p)$ in the fiber over p. In machine learning, a section is exactly the formal counterpart of a "feature map on a manifold."

4 Gauge Transformations

Definition 4.1: Gauge Transformation

A **gauge transformation** is a smooth choice of group element $g(p) \in G$ at each point $p \in \mathcal{M}$, varying continuously with p. It changes the local frame at every point independently:

$$e_i(p) \mapsto \sum_j g(p)_{ji}\, e_j(p).$$

The collection of all such choices forms the *gauge group* $\mathcal{G} = C^\infty(\mathcal{M}, G)$.

In physics, changing the gauge corresponds to changing the units or reference frame of measurement — it should not affect observable quantities. In ML, the gauge freedom corresponds to the *arbitrary choice of local coordinate system* used to represent features at each point.

Definition 4.2: Gauge Equivariance

A map f between feature spaces is **gauge-equivariant** if, under any gauge transformation $g(\cdot)$, it transforms as

$$f[g \cdot \text{input}] = g \cdot f[\text{input}],$$

where the action of g on the input and output features is given by the representation of G on the respective fibers.

The key insight: a physically (or geometrically) meaningful computation must be gauge-equivariant — its predictions cannot depend on our arbitrary choice of local reference frame.

Remark 4.1: Global vs. gauge equivariance

It is worth contrasting the two notions side by side. A *globally equivariant* model transforms a single, uniform group element across the entire input. A *gauge-equivariant* model must respect group elements that can vary continuously across the manifold — an infinite-dimensional family of transformations. Gauge equivariance is therefore a *strictly stronger* constraint whenever the base manifold is non-trivial, leading to even sparser layers and stronger inductive bias.

5 Connections and Parallel Transport

A central problem when working on manifolds is: *how do we compare feature vectors at different points?*

On flat $\mathbb{R}^d$ this is trivial — vectors at different points live in the same space and can be directly added or compared. On a curved manifold, the tangent spaces $T_p\mathcal{M}$ and $T_q\mathcal{M}$ are different vector spaces; there is no canonical way to "move" a vector from p to q.

Definition 5.1: Connection (intuitive)

A **connection** ∇ on a fiber bundle specifies, for each path γ from p to q in $\mathcal{M}$, a linear isomorphism

$$\Pi_\gamma : F_p \xrightarrow{\sim} F_q$$

called **parallel transport**. It tells us how to "slide" a feature vector along the manifold in a consistent way.

Example 5.1: Parallel transport on the sphere

Start at the North Pole with a tangent vector pointing East. Transport it southward along the Greenwich meridian, then East along the equator, then northward back to the starting point. The returned vector is *rotated* relative to the original one — by an angle equal to the solid angle enclosed by the path. This *holonomy* is a direct measure of the manifold's curvature.

Definition 5.2: Holonomy and curvature

The **holonomy** along a closed loop γ at p is the group element $\Pi_\gamma \in G$ obtained by parallel-transporting around γ. For an *infinitesimal* loop, the holonomy is governed by the **curvature** of the connection. On a flat manifold, all holonomies are the identity; non-trivial holonomy detects curvature.

Remark 5.1: Gauge and connection in neural networks

In a gauge-equivariant neural network on a manifold, the connection is used in the *message-passing* step: before aggregating feature vectors from neighbouring nodes, each neighbour's feature is first *parallel-transported* to the central node's local frame. Without this correction, the aggregation would mix features expressed in different (incompatible) coordinate systems.

6 From Physics to Machine Learning

The ideas above are not merely abstract: they directly inspire concrete neural network architectures.

6.1 Gauge Equivariant Mesh CNN

Cohen et al. (2019) introduced *Gauge Equivariant Mesh CNNs* for processing data defined on arbitrary triangle meshes (e.g. 3D shapes in computer graphics and medical imaging). Each node has a local frame; the convolution kernel is expressed in this frame; the parallel transport connection is used to relate frames at neighbouring nodes. The result is a network whose output is independent of the (arbitrary) choice of local frame on each triangle.

6.2 Equivariant Networks on Spheres

Weather forecasting, climate modelling, and cosmology involve data on S^2. Standard CNNs distort the data because the sphere cannot be unrolled flat. Sphere-equivariant networks (e.g. Cohen et al. 2018, *Spherical CNNs*) use the geometry of S^2 directly and respect its symmetry group SO(3).

6.3 Connection to Attention Mechanisms

A recent line of research (e.g. *Transformer on Riemannian Manifolds*) interprets attention as a form of parallel transport: the attention scores weight how much information to move from each neighbour, while value projections can be seen as transport operators in a feature bundle. This perspective suggests principled ways to extend transformers to non-Euclidean domains.

6.4 Geometric Algebra Networks

A further development is the use of *Clifford algebras* and *geometric algebra* as native representations for equivariant features. Networks such as *Clifford Algebra Networks* (Brandstetter et al. 2023) represent features as multivectors that transform under $\mathrm{O}(p, q)$ in a coordinate-free way, removing the need to explicitly track frames at all. This approach has shown promise for physics-based simulation, where electromagnetic and stress tensors are naturally multivector quantities.

7 The Coordinate Independence Principle

The central philosophical takeaway of this section is:

Coordinate Independence Principle. A geometrically meaningful operation should yield results that do not depend on the (arbitrary) choice of local reference frame. In mathematical terms: it must be *gauge-equivariant*.

This principle is the *non-Euclidean generalisation* of the equivariance principle of Section 1.2. When the domain is flat, gauge equivariance reduces to the global equivariance of standard geometric deep learning. On curved manifolds, it enforces a richer, locally varying symmetry that is the correct inductive bias for data living on surfaces and manifolds.

Exercises

1. Explain in your own words why a global rotation group such as SO(2) is insufficient to describe symmetries on the sphere S^2. What goes wrong at the poles?

2. A temperature field on the Earth's surface assigns a scalar $T(p) \in \mathbb{R}$ to each point p. Is T a section of the tangent bundle TS^2 or a section of the *trivial bundle* $S^2 \times \mathbb{R}$? What about a wind velocity field?

3. Describe qualitatively what happens if a message-passing neural network on a triangle mesh *ignores* the connection (i.e. aggregates neighbour features without parallel transport). In which situations would this cause the most error?

4. Sketch why the hairy ball theorem (Remark **??**) *does not* apply to the torus $T^2 = S^1 \times S^1$. What does this tell you about the existence of global frames on the torus?

5. Suppose two paths γ_1 and γ_2 both connect p to q on a manifold. Under what condition do they induce the same parallel transport map? (*Hint: consider the closed loop formed by going along γ_1 and back along γ_2.*)

6. (*Challenge*) Consider a fiber bundle over the circle S^1 with fiber $\mathbb{R}$. The *trivial bundle* is $S^1 \times \mathbb{R}$ (a cylinder). The *Möbius bundle* introduces a twist: as you go around S^1 once, the fiber undergoes a reflection $x \mapsto -x$. A gauge transformation at angle θ is a smooth choice of sign $g(\theta) \in \{+1, -1\}$. Can the Möbius bundle be made "trivial" by a gauge transformation? Justify your answer by considering what happens to a global non-vanishing section.